

A grayscale micrograph of a spinal cord cross-section. The image shows the characteristic butterfly shape of the gray matter, with the central canal visible as a small, dark, oval structure in the center. The surrounding white matter is lighter and has a more fibrous appearance. The overall structure is symmetrical and shows clear anatomical details.

# Spinal Cord

BIOL2251

Chapter 13

Full Chapter



# Lecture Quiz



**Test what you know!**



**9 questions** (MC + **Ordering**)

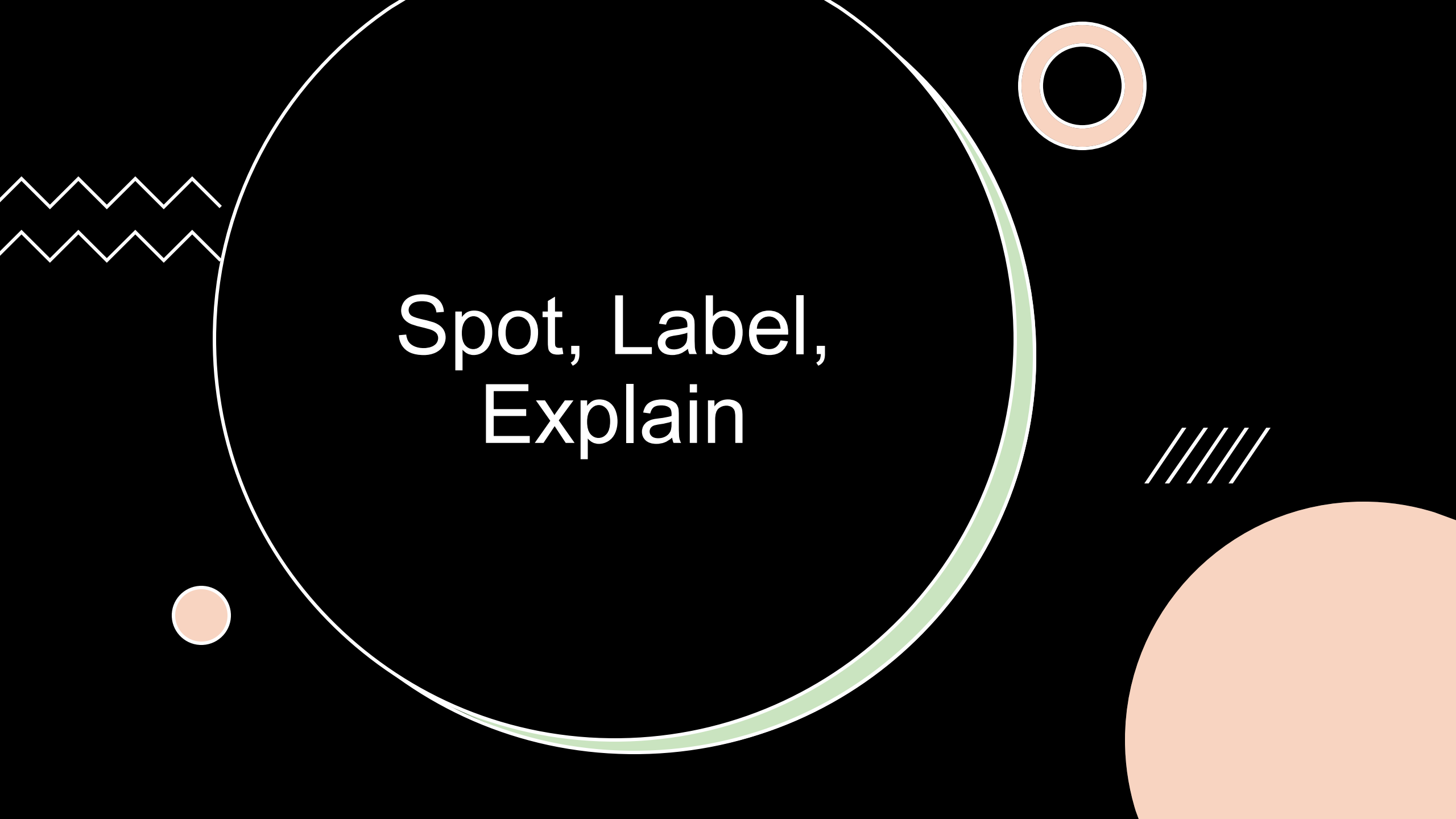


**Quiz automatically submits at 2:30 PM**



**Password = plexus**

**Make sure you sign in before the end of lecture!**

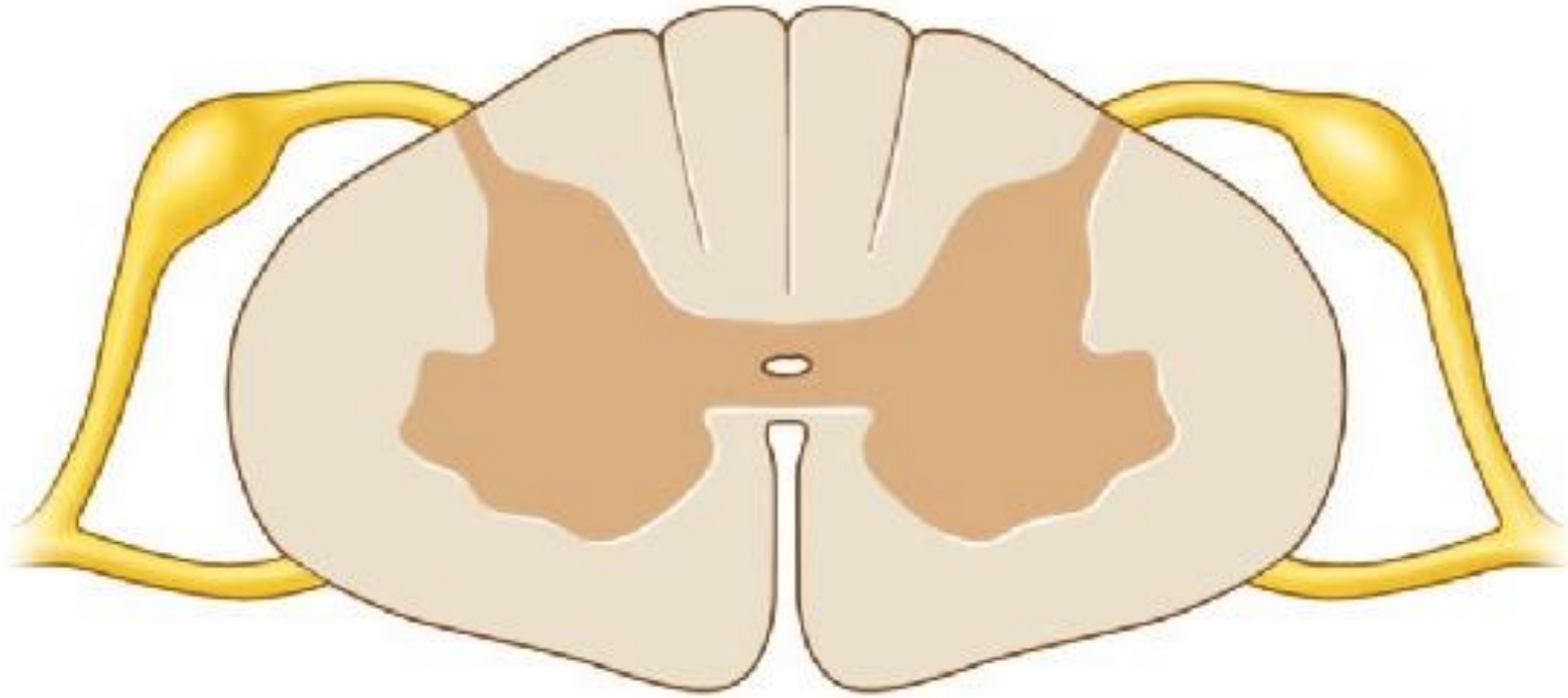


Spot, Label,  
Explain

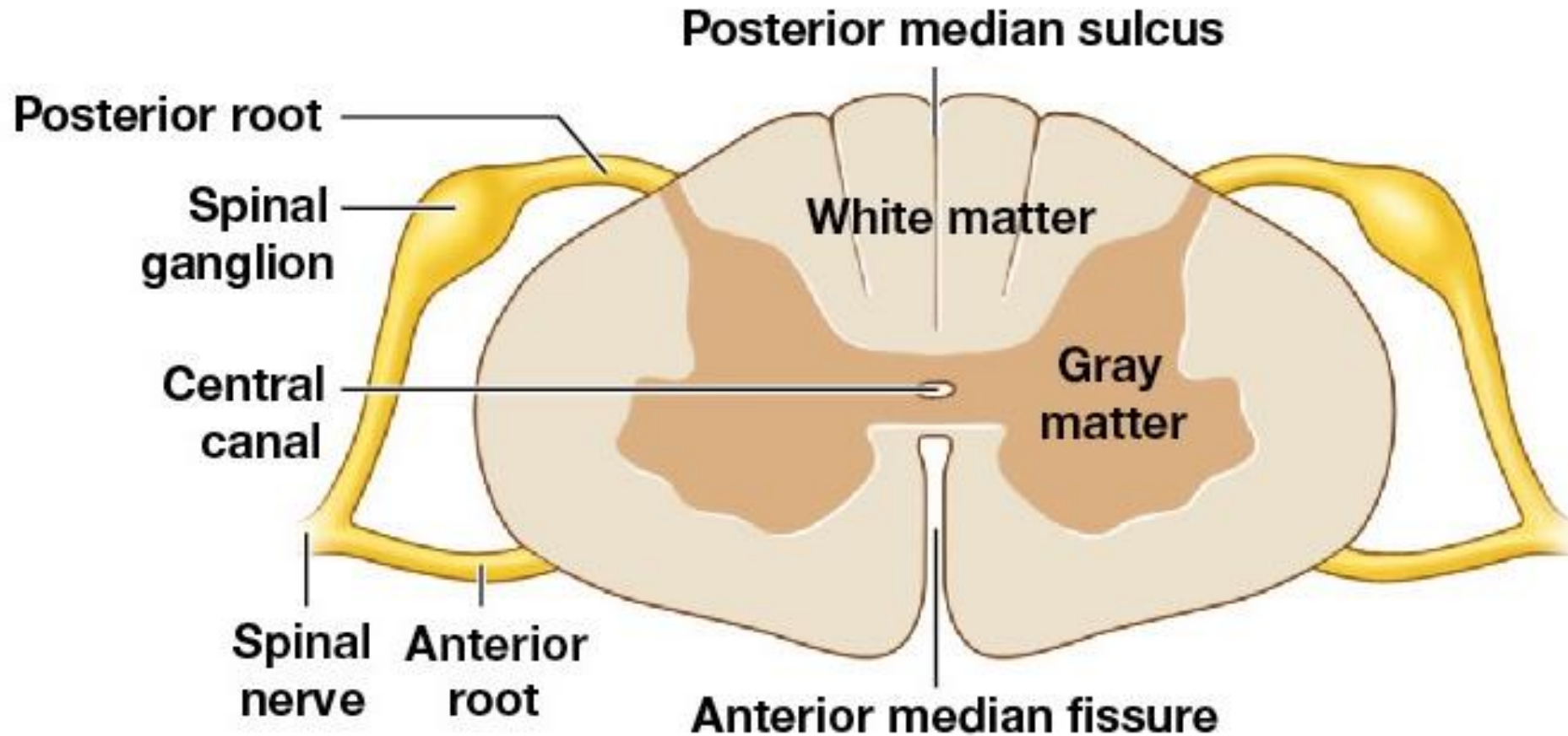
# Spinal Cord Anatomy

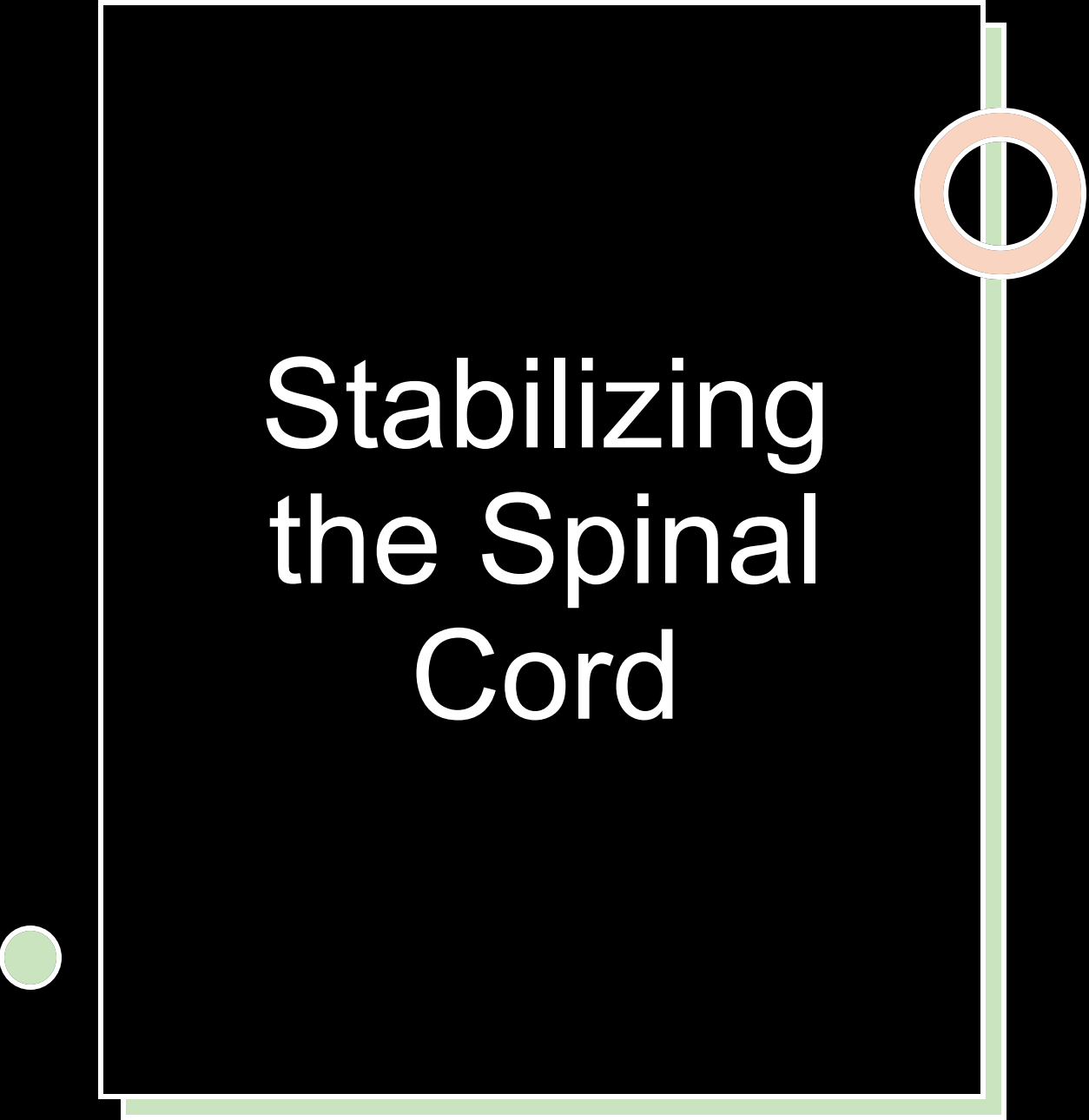
- **Instructions**
- On the projected diagram, label the following:
  - Ventral roots
  - Dorsal roots
  - Dorsal root ganglion
  - Anterior median fissure
  - Posterior median sulcus
- With a partner, compare labels.
- Together write a 1-sentence explanation: **Why are the cervical and lumbar regions enlarged?** Be ready to share.
- **Bloom's taxonomy = Remember/Understand**

# Spinal Cord Anatomy



# Spinal Cord Anatomy





# Stabilizing the Spinal Cord



# Stabilizing the Spinal Cord

- **Instructions**

- Work in teams of 4-6 (max of 12 groups)
- I will read a prompt about a **structure that stabilizes the spinal cord**
- For each prompt, your team will:
  - Write your answer on your whiteboard
  - Hold it up when ready
- Fastest correct answer earns a point!

- **Bloom's taxonomy = Understand/Apply**



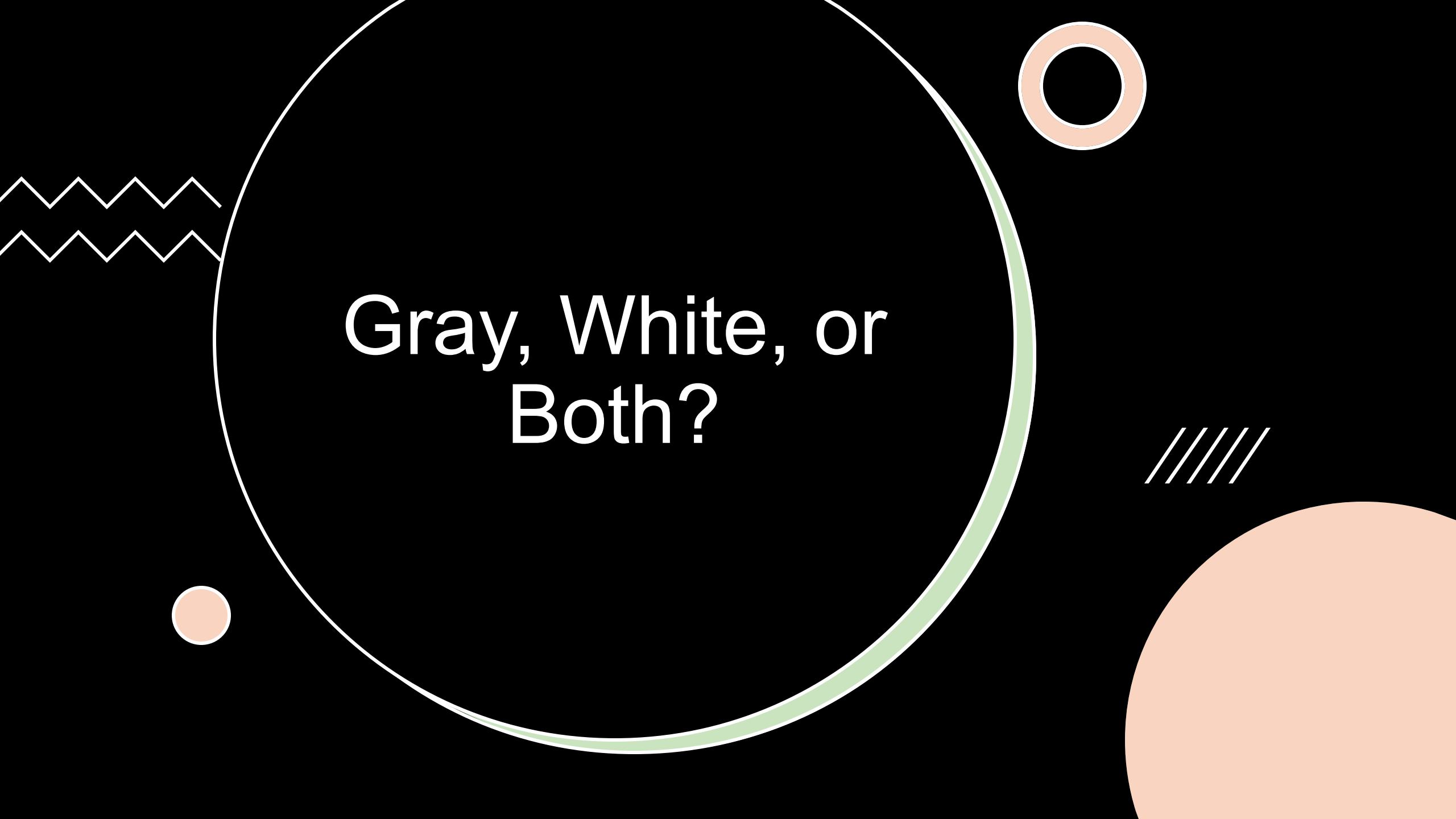
What prevents **lateral**  
movement of the spinal cord?

What prevents  
**superior/inferior** movement?

What prevents excessive  
**twisting** or **movement**  
generally?

# Stabilizing the Spinal Cord

- **Lateral movement:** denticulate ligaments
- **Superior/inferior movement:** connection of dura mater at foramen magnum+ coccygeal ligament
- **General stabilization:** meninges + CSF cushioning



Gray, White, or  
Both?



# Gray, White, or Both?

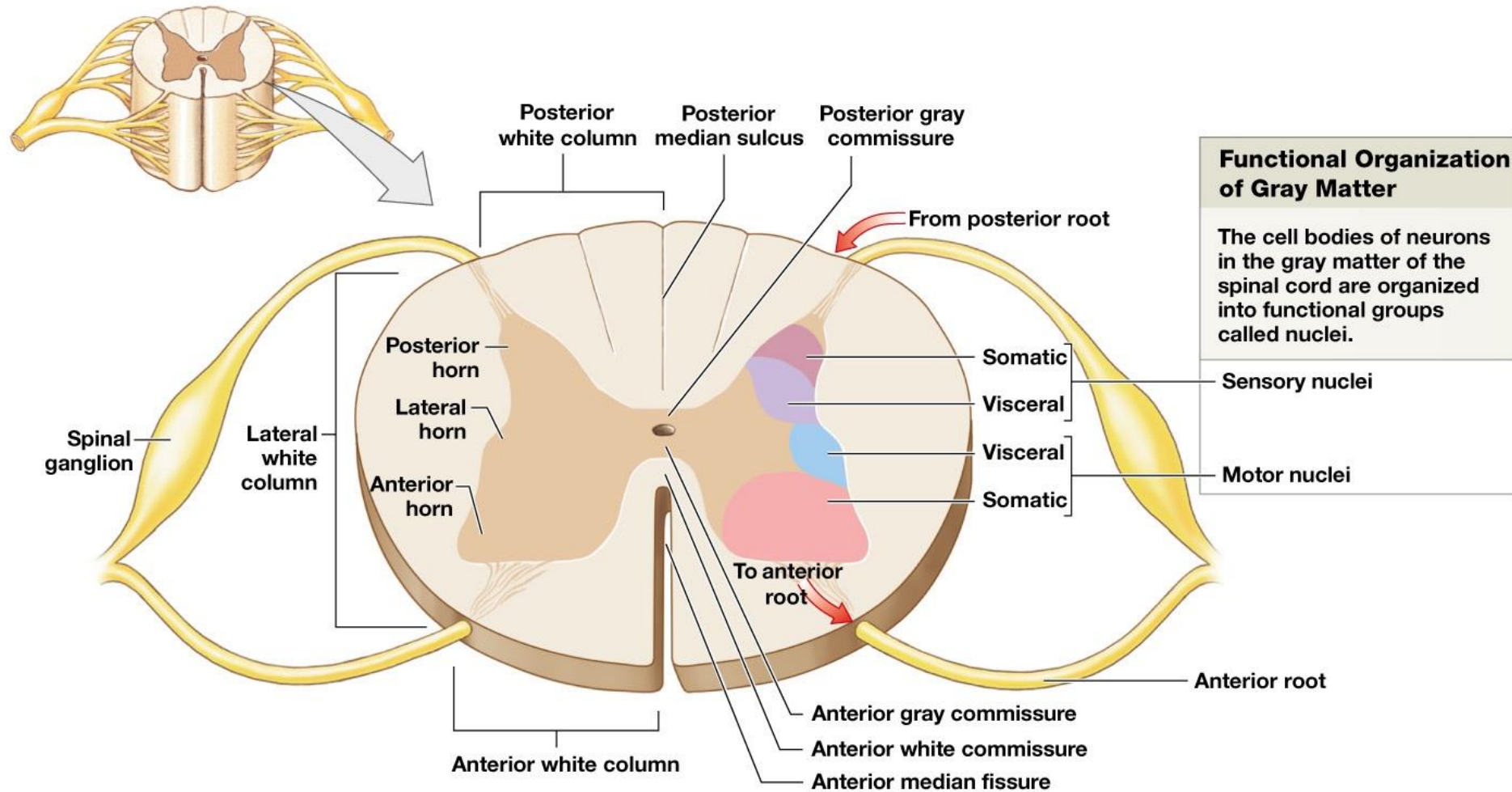
- **Instructions**

- Work in groups of 4
- Sort each term into **Gray Matter, White Matter, or Both:**
  - Anterior horn
  - Posterior horn
  - Lateral horn
  - Motor nuclei
  - Sensory nuclei
  - Anterior column
  - Posterior column
  - Anterior commissure
  - Posterior commissure
  - Ascending tracts
  - Descending tracts
- Your group will then answer two mini-cases.
- **Bloom's taxonomy = Understand/Apply/Analyze**

# Gray, White, or Both?

- **Gray Matter:** anterior horn, posterior horn, motor nuclei, sensory nuclei, anterior/posterior gray commissures
- **White Matter:** anterior/posterior columns, anterior/posterior white commissures, ascending and descending tracts
- **Both:** Anterior/posterior commissures (depending on specificity)
  - There are both white and gray commissures (anterior/posterior)

# Gray, White, or Both?



# Gray, White, or Both? (Mini-Cases)

- **Case 1:** “A patient has lost fine touch and vibration sensation below T6. Which spinal cord structure is likely affected?”
  - **Model Answer:** Posterior root, posterior horn of the gray matter, and/or ascending tracts.
- **Case 2:** “A patient cannot move their right leg, but sensation is intact. Which spinal cord structure is likely damaged?”
  - **Model Answer:** Anterior root, anterior horn of the gray matter, and/or descending tracts of the white matter.



# Reflex Arc Build

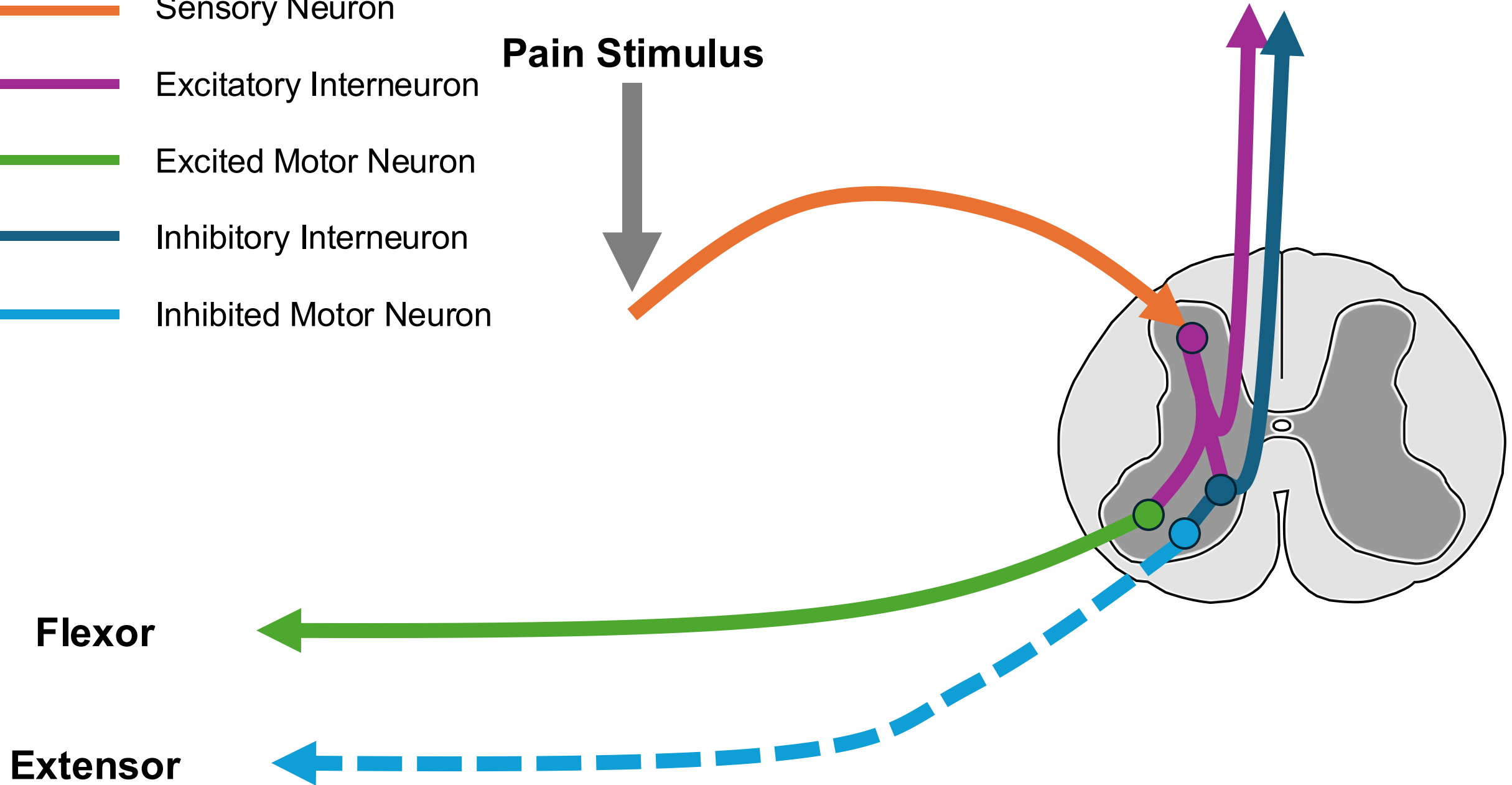
# Reflex Arc Build

- **Instructions**

- Work in groups of 4
- Reconstruct the correct order of events for:
  - **Flexor reflex (monosynaptic)**
  - **Crossed extensor reflex (polysynaptic)**
- **Bloom's taxonomy = Apply/Analyze/Evaluate**

-  Sensory Neuron
-  Excitatory Interneuron
-  Excited Motor Neuron
-  Inhibitory Interneuron
-  Inhibited Motor Neuron

**Pain Stimulus**





# Steps in the Flexor Reflex

1. Painful synapse is transmitted to sensory neuron on dorsal side of the spinal cord
2. Sensory neuron synapses onto an excitatory interneuron in the spinal cord
3. Excitatory interneuron stimulates an excitatory motor neuron, which signals a flexor to contract and pull away from the painful stimulus
4. Excitatory interneuron also stimulates an inhibitory interneuron
5. Inhibitory interneuron synapses with the motor neuron that inhibits the extensor (reciprocal inhibition)
6. Both the excitatory and inhibitory interneurons send information to the brain so the person is aware of what happened

- Sensory Neuron
- Excitatory Interneuron
- Excited Motor Neuron
- Inhibitory Interneuron
- Inhibited Motor Neuron

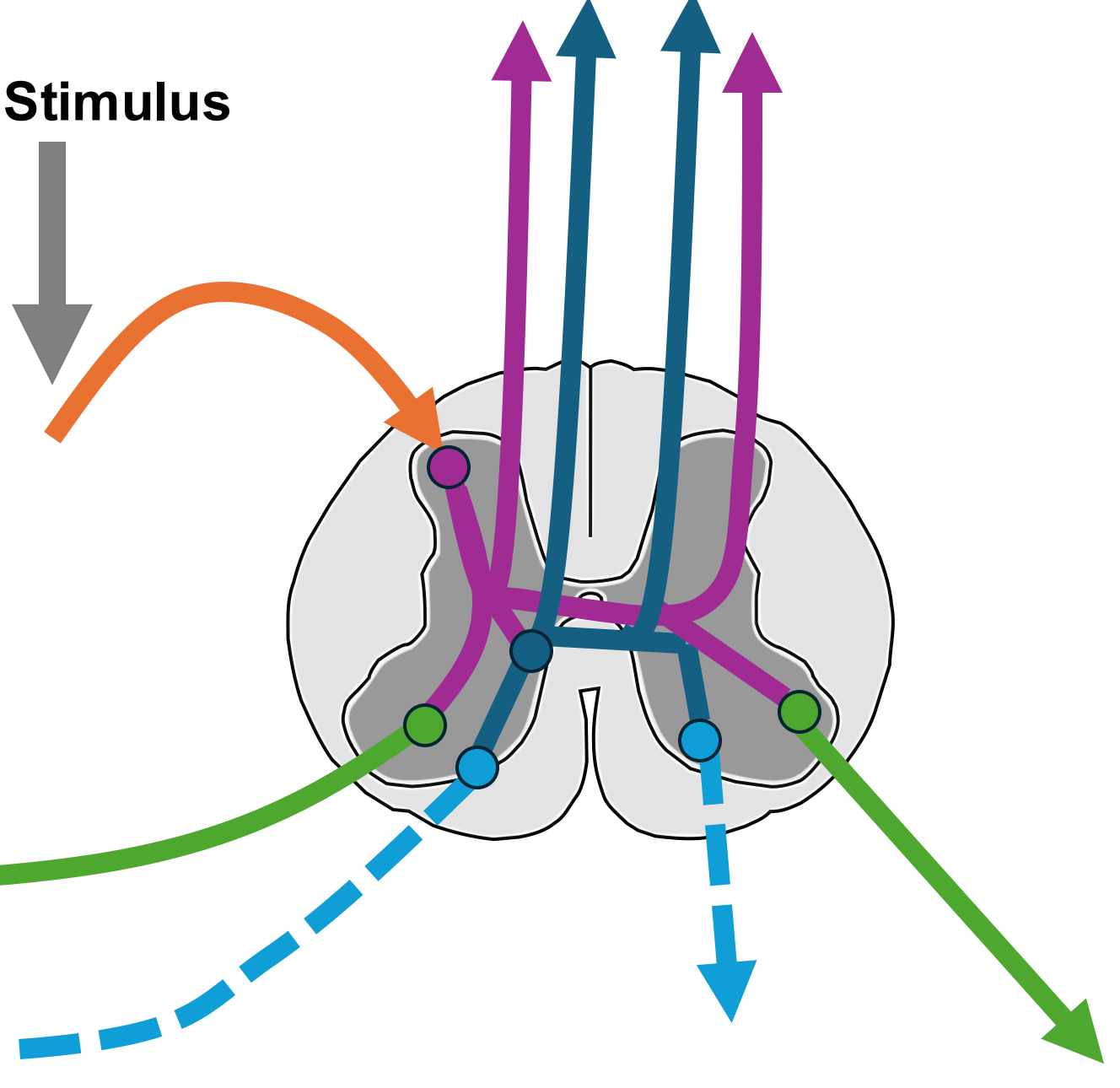
**Pain Stimulus**

**Flexor**

**Extensor**

**Flexor**

**Extensor**



# Steps in the Crossed Extensor Reflex

1. Painful synapse is transmitted to sensory neuron on dorsal side of the spinal cord
2. Sensory neuron synapses onto an excitatory interneuron in the spinal cord
3. Excitatory interneuron stimulates an excitatory motor neuron, which signals an ipsilateral flexor to contract **and a contralateral extensor to contract**
4. Excitatory interneuron also stimulates an inhibitory interneuron
5. Inhibitory interneuron synapses with the motor neuron that inhibits the ipsilateral extensor (reciprocal inhibition) **and inhibits the contralateral flexor**
6. Both the excitatory and inhibitory interneurons send information to the brain so the person is aware of what happened

# Reflex Arc Build (Part B)

- **Instructions**

- In your same groups, answer the following cases:
  - **Case 1:** A person steps on a tack with the left foot. Which limb flexes? Which limb extends?
  - **Case 2:** A patient reports numbness on the lateral forearm and weak elbow flexion. Which plexus and which dermatome/nerve are likely involved?

- **Bloom's taxonomy = Apply/Analyze/Evaluate**

# Reflex Arc Build: Case 1

- **Case 1:** A person steps on a tack with the left foot. Which limb flexes? Which limb extends?
- Left limb flexes (withdraws); **right limb extends** (crossed extensor reflex).

# Reflex Arc Build: Case 2

- **Case 2:** A patient reports numbness on the lateral forearm and weak elbow flexion. Which plexus and which dermatome/nerve are likely involved?
- **Brachial plexus (C5–C6) involvement; musculocutaneous nerve; dermatome C6 (lateral forearm).**

